

Revision Plan: Integrating the Robust-Evaluation Experiments

Paper: *Business Location Recommendation with Geospatial Data and Generative AI* (Kakepakis, Houstis, Vavalis) **Input:** `geospatial_llm_business_location_robust_evaluation.ipynb` + collaborator's real-data runs (Sections 4, 5, 7, 8, 9) **Purpose:** map each new experiment to the specific review criticism it answers, and give paste-ready text, table skeletons, and placement instructions.

0. Read this first — the one caveat that governs everything

The notebook is a *robust-evaluation framework*. Every number saved inside it (e.g. `NDCG@10 = 0.51`, `CSR@3 = 0.71`, `Jaccard = 0.70`) comes from a **synthetic, Volos-like testbed** and is illustrative only. Your collaborator re-ran the relevant sections on the **real** dataset and reported the *direction* of each finding, not the figures.

Therefore, throughout this plan:

- Draft prose states the **qualitative conclusion** your collaborator reported (which is what carries the scientific argument).
- Every specific figure is written as a `{placeholder}` to be filled from the real run.
- Section 5 lists exactly which real numbers/figures to request.

Do **not** paste the notebook's synthetic numbers into the paper. They are scaffolding, not results.

1. The mapping: which observation answers which criticism

Review criticism (from prior critique)	Collaborator experiment (notebook section)	What it establishes
Circular proxy-target construction — LLMs may only be re-learning a proxy built from the same spatial signals	\$4 baselines + \$9 ablation	Classical CF / content-based baselines reach comparable top-N accuracy without an LLM; presence is mostly explained by centrality, so "accuracy on the proxy" is not the LLM's real contribution
What does the LLM actually add?	\$4 content-based RF (category-encoding vs description-embedding) + \$5 constraint satisfaction	Embeddings do not improve raw recommendation; the LLM's measurable value is constraint extraction , quantified by CSR@3
Precision@3 = Recall@3 degeneracy	\$4 interaction-grounded leave-one-out	Held-out empirical interactions of variable size make Precision and Recall genuinely distinct, replacing the equal-set-size artifact
Misleading explainability framing / SHAP collinearity	\$9 ablation ("which features relate to business presence?")	Presence is dominated by centrality; per-class feature "variety" is partly cosmetic — this reframes the justification claims honestly
Arbitrary proxy weights (0.311 / 0.358 / 0.331)	\$7 proxy sensitivity	Top-10 proxy sets are stable across equal / dominant / random weightings — the proxy is not an artifact of one hand-tuned weighting
Arbitrary 48-Voronoi partition / generalizability	\$8 spatial robustness	Conclusions persist when neighborhoods are merged into coarser zones (16/24/32)

The unifying message (this is the spine of the revision): the paper's original metrics measure *emulation of a proxy*; the new experiments are an *independent* test, and they relocate the LLM's contribution from raw accuracy to natural-language constraint handling. This is exactly your collaborator's conclusion, and it turns three of the review's sharpest criticisms into deliberate, defended design choices.

2. Section-by-section modifications

For each block: **(a)** where it goes, **(b)** the criticism it answers, **(c)** paste-ready draft text, **(d)** a LaTeX table/figure skeleton, **(e)** the real numbers to plug in.

2.A — Interaction-grounded baselines & the embedding-vs-encoding test

(a) Placement. New subsection **5.4 "Interaction-Grounded Baselines"**, immediately after the Table 3/4 comparison. This also lets you promote the implicit-feedback material currently sitting in *Future Work* (Fig. 5) into the main evaluation, since it is no longer "preliminary."

(b) Answers. Circularity; "what does the LLM add"; the Precision@3 = Recall@3 degeneracy.

(c) Draft text.

To test whether the LLM architectures contribute beyond emulating the proxy ensemble, we evaluate them against standard recommenders under an *interaction-grounded* protocol. The observed presence of a NACE-class business in a neighborhood is treated as an implicit positive; for each class one observed interaction is held out, and models rank the remaining neighborhoods. Because the held-out set has variable size, Precision and Recall are no longer forced to coincide as they are under the equal-cardinality emulation metrics of Table 3.

We compare three families: (i) collaborative-filtering baselines (TopPop, ItemCF, PureSVD, a NeuMF-style model); (ii) content-based rankers — a Random Forest over neighborhood features, which is content-based rather than collaborative because it scores (class,

neighborhood) pairs from side information rather than co-occurrence; and (iii) the proposed LLM pipelines. To isolate the effect of richer category representations, the content-based ranker is run in two variants: one encoding the business category with a simple categorical encoding, and one replacing that encoding with a dense embedding of the category description.

Two findings stand out. First, classical collaborative-filtering and content-based rankers attain top-N accuracy comparable to — and on the long-tail subset competitive with — the LLM pipelines, confirming that raw recommendation accuracy on the proxy is not where the generative models add value. Second, substituting description embeddings for the simple encoding does **not** improve recommendation quality ($\Delta\text{NDCG}@10 \approx \dots$), indicating that the gains attributable to language models do not arise at the retrieval/ranking stage. We return in Section 5.5 to where the LLM contribution does materialise.

(d) LaTeX skeleton.

```
\begin{table}[t]
\centering
\caption{Interaction-grounded leave-one-out evaluation (real Volos data).
Held-out empirical co-locations as targets; Precision and Recall are
non-degenerate. CF = collaborative filtering; CB = content-based.}
\label{tab:interaction_baselines}
\begin{tabular}{lcccc}
\hline
Method & HR@10 & NDCG@10 & Prec@10 & Rec@10 \\
\hline
TopPop (CF) & & & & \\
ItemCF (CF) & & & & \\
PureSVD (CF) & & & & \\
NeuMF-style (CF) & & & & \\
RF, simple encoding (CB) & & & & \\
RF, desc.\ embedding (CB) & & & & \\
\hline
Fine-tuned LLM & & & & \\
RAG pipeline & & & & \\
\hline
\end{tabular}
\end{table}
```

(e) Fill from real data. HR@10 / NDCG@10 / Prec@10 / Rec@10 for each method on the full test set; the RF simple-encoding vs embedding pair (and the ΔNDCG gap to quote inline); optionally the long-tail subset numbers if you keep the full/long-tail split from Fig. 5.

Note on the existing Fig. 5. Your paper already shows CF baselines (HR/NDCG, full vs long-tail) in *Future Work*. Move that figure here and cite it from 5.4, so the CF comparison becomes a main result rather than a forward-looking sketch. The embedding-vs-encoding rows are the genuinely new content.

2.B — Constraint satisfaction (where the LLM actually wins)

- (a) **Placement.** New subsection 5.5 "Constraint Satisfaction", right after 5.4. Add one column to Table 4.
- (b) **Answers.** "What does the LLM add"; replaces the single anecdote (the port → Palaia example on p.12) with a metric; supports the *Personalization* row of Table 4.
- (c) **Draft text.**

The comparison above shows that pure recommenders match the LLM pipelines on proxy accuracy, but it does not capture the conversational requirement that motivates a generative interface: honoring free-text user constraints such as *near the port* or *in a quiet area*. We therefore evaluate **Constraint Satisfaction Rate at 3** (CSR@3): for each (class, constraint) query, the fraction of the top-3 recommendations that satisfy the stated constraint, where a constraint is operationalised as membership in the appropriate quantile of the relevant feature (e.g. nearest-30% distance to the port).

A purely interaction-driven recommender, lacking any channel to read the constraint, defaults to globally attractive — typically central — neighborhoods regardless of what the user asks for, achieving only CSR@3 = {...}. Introducing the LLM layer, which parses the constraint and re-ranks candidates accordingly, raises this to CSR@3 = {...}, satisfying nearly all stated constraints. This is the decisive, non-circular contribution of the generative component: it is not visible in proxy-emulation accuracy (Table 3) nor in interaction-grounded accuracy (Table 5/§5.4), but only when recommendations are judged against natural-language preferences that structural recommenders cannot represent.

(d) LaTeX skeleton.

```
\begin{table}[t]
\centering
\caption{Constraint Satisfaction Rate@3 (real Volos data). A pure recommender
cannot read free-text constraints; the LLM layer parses and enforces them.}
\label{tab:csr}
\begin{tabular}{lc}
\hline
System & CSR@3 \\
\hline
Pure recommender (CF / static proxy top-3) &  $\angle\angle$  \\
LLM-augmented (constraint-aware re-ranking) &  $\angle\angle$  \\
\hline
\end{tabular}
\end{table}
```

Then add a **Constraint satisfaction** row to Table 4: Fine-Tuned = *Limited*, RAG/LLM-augmented = *High*.

(e) Fill from real data. The two CSR@3 numbers; optionally a per-constraint breakdown (port / university / center / quiet / main-road / transport) if your collaborator logged it — that makes a strong supplementary bar chart.

2.C — Proxy weight sensitivity

(a) Placement. Add a short paragraph to **Section 3.3 (Proxy Ground Truth Creation)**, right after Eq. (1) and the weight values.

(b) Answers. The arbitrariness of the specific weights (0.311 / 0.358 / 0.331).

(c) Draft text.

Because these weights are derived rather than learned, we verify that the resulting targets are not an artifact of one particular weighting. Recomputing the proxy under equal weights, under each single-method-dominant weighting, and under 100 random weight vectors drawn from a Dirichlet prior, we measure the mean top-10 Jaccard overlap against the reference proxy. Overlap remains high across all scenarios (equal weights ($J \approx \dots$); mean over random weightings ($J \approx \dots \pm \dots$)), indicating that the consensus ranking is stable and dominated by agreement among the three methods rather than by the exact mixing coefficients.

(d) Optional small table.

```
\begin{table}[t]
\centering
\caption{Stability of the proxy under alternative rank-aggregation weights:
mean top-10 Jaccard vs.\ the reference proxy (real Volos data).}
\label{tab:proxy_robust}
\begin{tabular}{lc}
\hline
Weighting scenario & Mean Jaccard@10 \\
\hline
Equal &  $\angle\angle$  \\
Supervised-dominant &  $\angle\angle$  \\
Unsupervised-dominant &  $\angle\angle$  \\
Sparse-rating-dominant &  $\angle\angle$  \\
Random (100 draws, mean $\pm$ sd) &  $\angle\angle \pm \angle\angle$  \\
\hline
\end{tabular}
\end{table}
```

(e) Fill from real data. Jaccard for the four named scenarios + mean/sd over the random draws.

2.D — Spatial partition robustness

(a) Placement. Add a paragraph to **Section 3.3** (or a short **5.6**), and reinforce the region-agnostic claim in **Section 6 (Conclusions)**.

(b) Answers. Dependence on the arbitrary 48-Voronoi partition; supports the "region-agnostic / replicable" claim.

(c) Draft text.

A second robustness concern is whether conclusions depend on the specific 48-cell Voronoi partition. We therefore re-aggregate neighborhoods into coarser spatial units ((16, 24, 32) regions, by clustering cell centroids) and repeat the ranking evaluation at each granularity. Recommendation quality (remains stable / degrades gracefully) as the partition coarsens ((report HR/NDCG trend)), and — importantly — the tendency of presence-driven scores to favour central units persists at every granularity. The substantive findings are thus a property of the underlying spatial signal, not of one partitioning choice.

(e) **Fill from real data.** HR/NDCG (or Match@3) at each granularity, and whether central-area dominance persists. State the actual direction; do not assume stability if the collaborator's recorded results differ.

2.E — Ablation: presence is mostly centrality (the honest reframe)

(a) **Placement.** Add to **Section 3.3** and revise the explainability language in **Section 4.1** and **Section 6**.

(b) **Answers.** Circularity (directly); misleading explainability framing; SHAP collinearity.

(c) **Draft text (for 3.3).**

An ablation over feature groups clarifies what the proxy actually encodes. Removing accessibility or demographic features individually leaves ranking quality largely intact, whereas the dominant explanatory factor for observed business presence is **centrality** — proximity to the city centre and to high-density cores. This is a deliberate caution rather than a flaw: it confirms that observed presence is a *centrality-biased* signal and explains why presence cannot be equated with suitability. It also tempers the interpretation of the per-class SHAP justifications: while the templated explanations cite class-specific features for linguistic variety, the ranking itself is driven mainly by a small set of mutually correlated centrality measures (consistent with the near-collinearity of centre/university/main-road distances reported in Section 3.2, r up to 0.99).

(c') **Revision to the explainability claim (Section 4.1 / abstract / conclusion).** Soften "evidence-based justifications" / "explainable" wherever it implies the cited features *cause* the ranking. Suggested replacement phrasing:

The SHAP-informed features provide *plausible, feature-grounded rationales* for the recommendations; we do not claim per-feature causal faithfulness, since an ablation analysis (Section 3.3) shows the rankings are driven primarily by correlated centrality measures.

(e) **Fill from real data.** The collaborator confirmed the centrality finding qualitatively; if they produce the feature-group ablation numbers, drop them into a small table mirroring 2.C/2.D. If not, the prose above stands on the qualitative result.

3. The narrative reframe (do this even if you add nothing else)

Two edits change how the whole paper reads and defuse the circularity criticism at the source:

1. **Name the original evaluation for what it is.** In Section 5 (and one line in the abstract), state that Match@3 / Precision@3 / Recall@3 / BERT-F1 measure *emulation* of the proxy ensemble — the capacity to reproduce its spatial logic — and that the interaction-grounded experiments of §5.4–5.5 provide the *independent* test. The paper already gestures at this on p.8 ("measure the capacity ... to replicate ... not to predict absolute commercial viability"); make it explicit and forward-reference the new sections.
2. **Relocate the headline claim.** The contribution sentence in the abstract and conclusion should move from "*the LLM gives accurate recommendations*" to "*classical recommenders match the LLM on proxy/interaction accuracy; the LLM's distinctive, measurable value is constraint extraction and natural-language personalization (CSR@3 of (...) vs (...)).*" This is both more defensible and more interesting, and it is precisely what the new experiments demonstrate.

4. Minor fixes flagged in review (quick wins)

- **Hardware contradiction.** The text says NVIDIA **A40** (Sections 4.1, 5, Table 3) but Fig. 1 / earlier notes say **A100**. Pick one (the runs appear to be A40) and make Fig. 1 consistent.
- **Precision@3 = Recall@3 in Table 3.** Add one sentence explaining the identity is forced by equal-cardinality sets (3 predictions vs a 3-item proxy target), and note that §5.4's variable-size held-out protocol removes the degeneracy. This converts a "bug-looking" artifact into a documented property.
- **GitHub URL.** The *Supplementary information* paragraph references a repository but the link is missing/placeholder. Insert the actual URL.
- **Broken citation.** Resolve the broken reference flagged in review (verify the [38] industry-report citation, which is out of sequence with the [1]–[27] list).
- **Forward-dated references.** Several entries are dated 2026 (refs 11, 12, 13, 19) and one is "submitted to ICLR 2026". Confirm these are intended (preprints/accepted) and consistent with the submission venue's policy.

5. What to request from your collaborator

To turn every (placeholder) above into a finalized result, ask for the **real-data outputs** of the sections they ran:

1. **§4 baselines table** — HR@10, NDCG@10, Precision@10, Recall@10 for TopPop, ItemCF, PureSVD, NeuMF, the two content-based RF variants (simple encoding vs description embedding), and the two LLM systems; full set and long-tail subset if available. (→ Table in 2.A; quote the embedding Δ NDCG.)
2. **§5 constraint satisfaction** — the two CSR@3 values, ideally with the per-constraint breakdown. (→ Table in 2.B.)
3. **§7 proxy sensitivity** — Jaccard@10 for the four named weightings + mean/sd over the 100 random draws. (→ Table in 2.C.)
4. **§8 spatial robustness** — HR/NDCG (or Match@3) at each granularity, and whether central-area dominance persists. (→ 2.D — and tell them the direction matters, not just the headline.)

5. **§9 ablation** — if obtainable, the feature-group ablation numbers behind the "presence $\approx$ centrality" conclusion. (→ 2.E; prose works without them.)

Also confirm two things that affect framing: (i) the dataset/run uses the **same real Volos data** as the main paper (so the baselines are directly comparable to Table 3), and (ii) whether the constraint masks were validated against actual geographic positions rather than the synthetic quantile proxy used in the notebook.

6. Suggested order of work

1. Apply the **narrative reframe** (Section 3) and **minor fixes** (Section 4) — these need no new numbers.
2. Promote the existing Fig. 5 into **5.4** and add the embedding-vs-encoding rows once the §4 table arrives.
3. Add **5.5 CSR@3** — this is the highest-impact new result; it is what justifies the LLM at all.
4. Fold proxy and spatial robustness (**2.C/2.D**) into Section 3.3 as two short paragraphs.
5. Revise the explainability claims (**2.E**) last, once the ablation framing is settled.